Using JetCam

Using JetCam
1. JetCam Installation
2. Using JetCam
2.1. CSI Camera
Main Code Explanation
Calling the Camera
Getting the Camera Image
2.1.1. Single-Channel Camera
2.1.2. Multi-Channel Camera
2.2. USB Camera
References

> **pip Installation Note:** Ubuntu 24.04 marks the system Python as "externally managed" according to PEP 668, so installing directly with pip may report `externally-managed-environment`. `--break-system-packages` allows pip to install packages in that environment; a virtual environment usually does not need this parameter, but this tutorial keeps it to keep the commands consistent.

> The factory system already has the relevant environment set up, so the installation steps can be skipped.

JetCam is an easy-to-use Python library developed by NVIDIA for the Jetson platform, used to integrate and operate USB cameras or CSI cameras

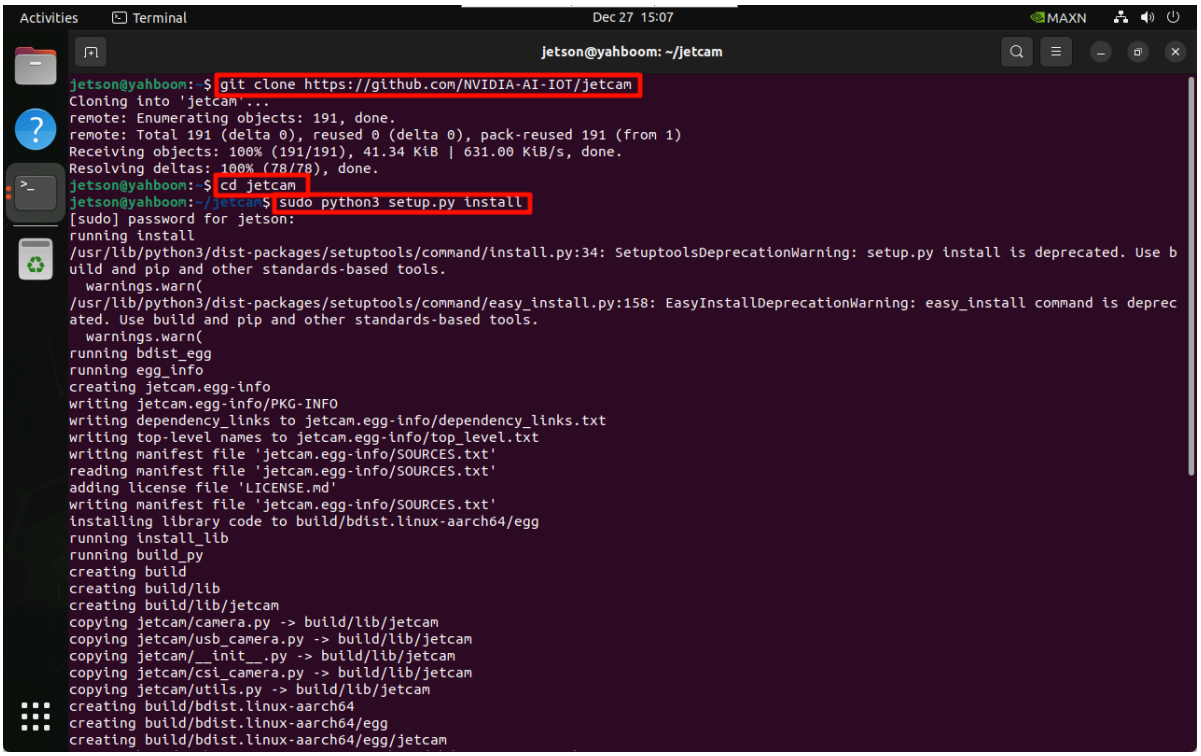
1. JetCam Installation

```
git clone https://github.com/NVIDIA-AI-IOT/jetcam
```

```
cd jetcam
```

```
sudo python3 setup.py install
```

```
sudo pip3 install ipywidgets --break-system-packages
```



2. Using JetCam

JetCam provides typical sample programs to demonstrate the calling of CSI and USB cameras to users.

```
The case needs to be run with Jupyter Lab. With the factory image system, you can access it directly through the mainboard IP:8888!
```

2.1. CSI Camera

In the Jupyter Lab web page, go to the folder where the CSI camera is located and open the corresponding folder:

```
/home/jetson/jetcam/notebooks/csi_camera
```

Note: If you are not very familiar with Jupyter Lab, you can read the Using Jupyter Lab tutorial to learn the basic operations!

Main Code Explanation

Calling the Camera

width: image output width

height: image output height

```
from jetcam.csi_camera import CSICamera
camera = CSICamera(width=224, height=224)
```

Getting the Camera Image

```
image = camera.read()
```

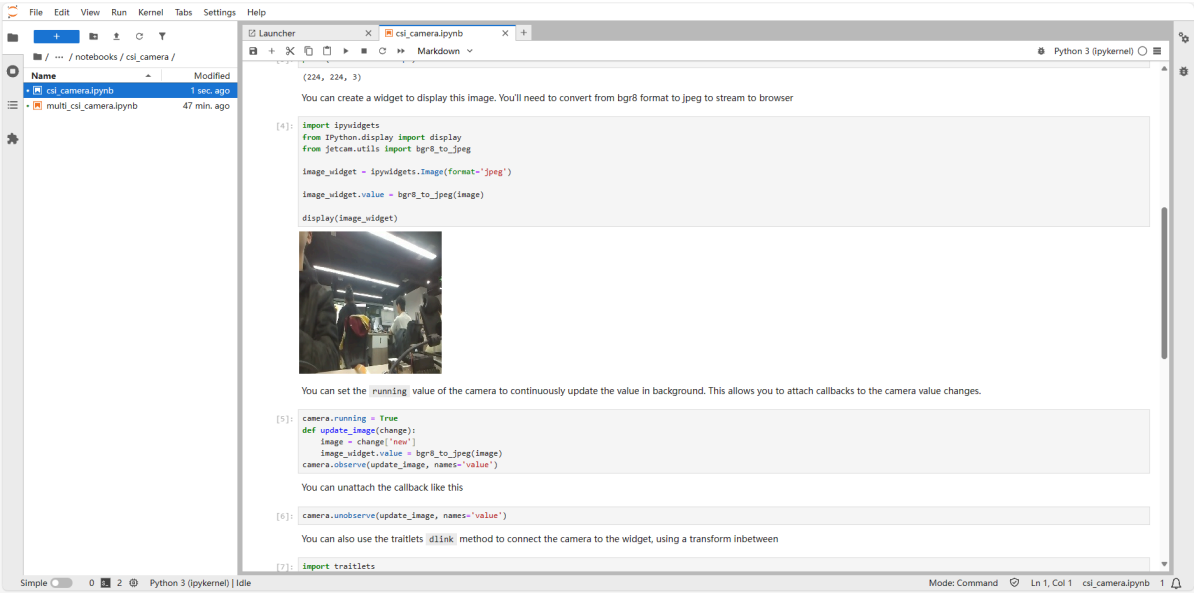
2.1.1. Single-Channel Camera

> Source Code Path

```
/home/jetson/jetcam/notebooks/csi_camera/csi_camera.ipynb
```

> Running Phenomenon

After the program file is opened, run the cells one by one from top to bottom:



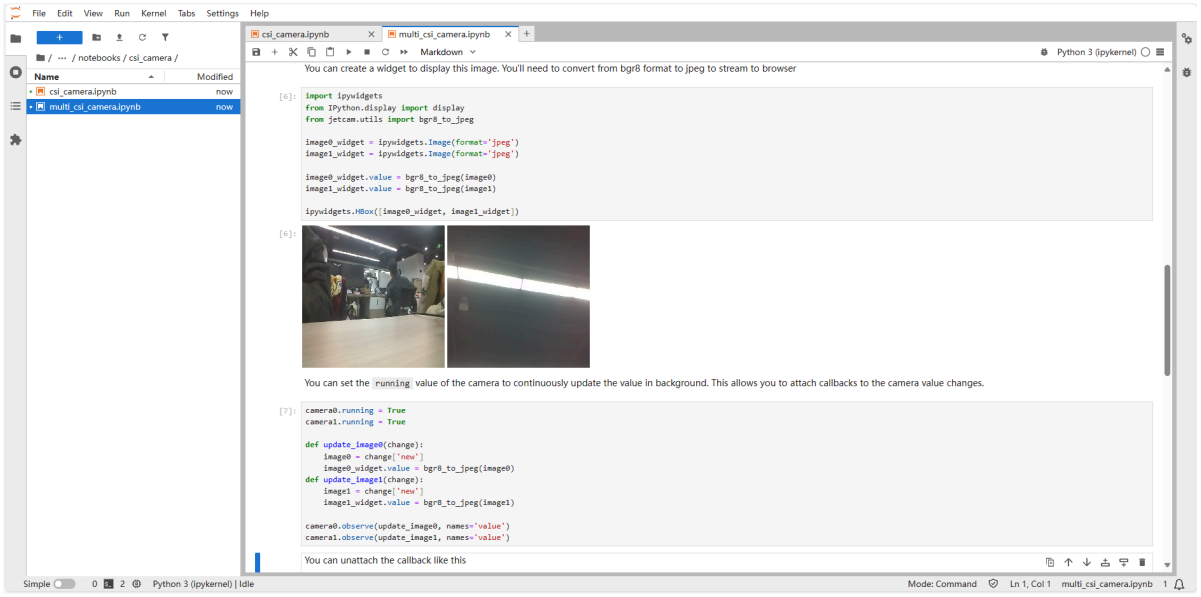
2.1.2. Multi-Channel Camera

> Source Code Path

```
/home/jetson/jetcam/notebooks/csi_camera/multi_csi_camera.ipynb
```

> Running Phenomenon

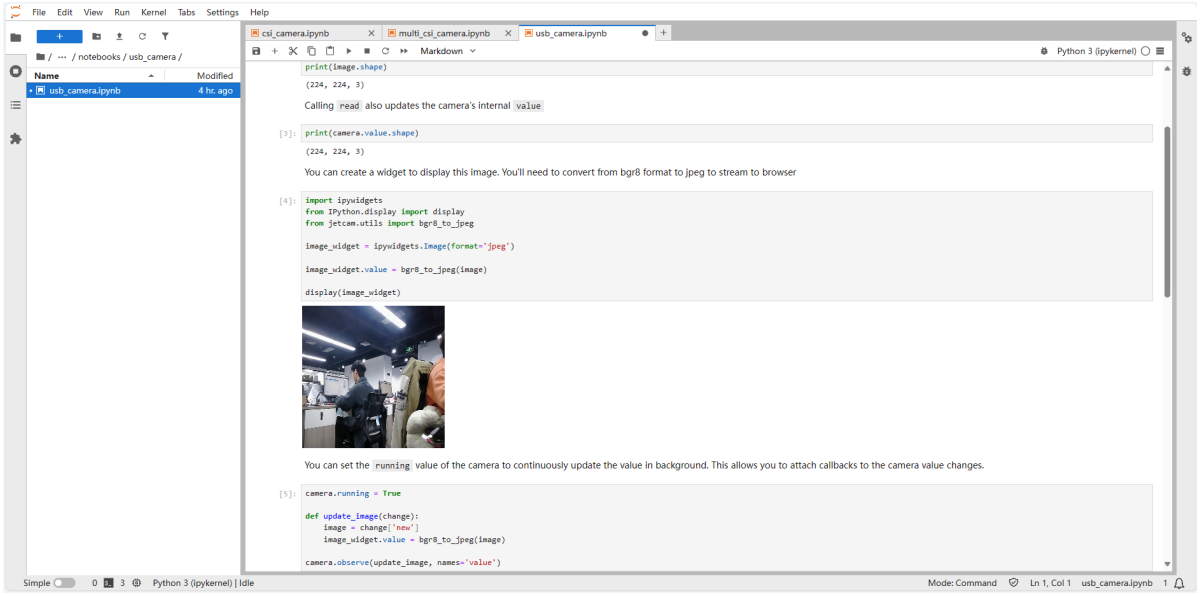
After the program file is opened, run the cells one by one from top to bottom:



2.2. USB Camera

In Jupyter Lab, go to the folder where the USB camera is located and open the file. The folder path in the factory image system is:

```
/home/jetson/jetcam/notebooks/usb_camera
```



References

https://github.com/NVIDIA-AI-IOT/jetcam